Hybrid U-Net and Transformer Models for 2D Single-Cell Segmentation

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***Abstract*—This project evaluates several deep learning architectures for 2D single-cell segmentation using phase-contrast microscopy images. We compare three models—a residual attention U-Net, a hybrid version with a lightweight Transformer bottleneck, and a deeper Transformer-based U-Net—to understand how self-attention and network depth affect segmentation quality. All models were trained on a publicly available dataset and assessed using Dice, IoU, precision, and recall. The hybrid U-Net with a Transformer bottleneck achieved the best overall performance, with a test Dice score near 0.90. These results suggest that adding limited global self-attention improves segmentation accuracy, while deeper Transformer modules provide diminishing returns for this dataset. The study highlights the effectiveness of hybrid architectures as an efficient approach for cell segmentation tasks.**

I. Introduction

Cell segmentation is an important step in analyzing biological images because it helps scientists identify and study individual cells in a sample. Doing this by hand can take a long time and is often inconsistent, so automated image analysis methods are becoming more popular. In this project, we focus on 2D cell segmentation using microscopy images of baby hamster kidney cell cultures.

Widefield fluorescence microscopy produces detailed images of cells, but it also introduces challenges such as uneven lighting, blurry edges, and overlapping cells. These factors make it difficult for simple image processing techniques to accurately separate one cell from another. To address this problem, we are building a neural network model using a Kaggle notebook to automatically separate the cells in these images.

The goal of this study is to create and test a model that can reliably identify individual cells in the microscopy images. By improving segmentation accuracy, this project aims to make cell image analysis faster, more consistent, and easier to apply in future biological research.

II. Methods

A. Dataset

We used the `SingleCellSegmentation` dataset of Guo et al. (Kaggle), which contains phase-contrast microscopy images with corresponding binary masks delineating cell pixels. We used the 512×512 version with the authors' original split: 4,140 training, 1,380 validation, and 2,070 test images.

B. Preprocessing

Images were decoded as RGB, converted to `float32`, and normalized to the $[0, 1]$ range. Masks were decoded as single-channel images and binarized (values > 127 mapped to 1, others to 0). Because all images were already 512×512 pixels, we did not apply any resizing, cropping, or geometric warping during preprocessing.

C. Data Augmentation

On the training set we applied simple, shape-preserving augmentations to reduce overfitting. Each image–mask pair was randomly transformed with horizontal and vertical flips (probability 0.5 each), brightness shifts (± 0.08), contrast scaling in $[0.9, 1.1]$, and additive Gaussian noise with standard deviation ≈ 0.03. Spatial transforms were applied identically to images and masks, while photometric transforms were applied only to images. Intensities were clipped back to $[0, 1]$.

D. Model Architectures

All models follow a standard encoder–decoder U-Net topology with skip connections. We focus on three variants:

- **Model A (ResAtt U-Net)**: a five-level U-Net backbone augmented with residual blocks and attention gates on the skip connections to improve gradient flow and suppress background features.
- **Model B (ResAtt U-Net + Transformer)**: identical to Model A except for a lightweight Transformer bottleneck at the 32×32 feature level. The encoder output is flattened into tokens, processed by several Transformer encoder blocks (multi-head self-attention and a two-layer feed-forward network), and reshaped back into a feature map before entering the decoder.
- **Deep Transformer U-Net**: a deeper variant with an additional encoder–decoder level and a larger Transformer bottleneck, increasing capacity and receptive field while maintaining the same overall U-Net structure.

E. Training Procedure

Models were implemented in TensorFlow/Keras and trained on Kaggle using two NVIDIA Tesla T4 GPUs with synchronous data-parallel training (MirroredStrategy). We used a two-stage schedule:

1) **Warmup stage**: a short phase optimized with a numerically robust binary cross-entropy (BCE) loss and the Adam optimizer (learning rate 5×10^{-5}) to obtain stable probability maps.
2) **Fine-tuning stage**: optimization with an edge-enhanced compound loss combining focal BCE, a global soft Dice term, and edge-aware terms that up-weight boundary pixels. Fine-tuning used AdamW (or Adam when AdamW was unavailable) with initial learning rate 2×10^{-4}, weight decay 1×10^{-4}, gradient clipping ($\|g\|_2 \leq 1.0$), and a cosine decay learning-rate schedule.

F. Evaluation Protocol

Hyperparameters were tuned on the validation set. At test time we used test-time augmentation (horizontal/vertical flips and 90° rotations) and averaged predictions across augmentations to obtain one probability map per image. For each model we swept the binarization threshold on the validation set and selected the value that maximized the hard Dice coefficient. Using this threshold, we evaluated on the held-out test set and reported hard Dice, Intersection over Union (IoU), pixel accuracy, precision, and recall.

III. Results

A. Overall Segmentation Performance

We evaluated Model A (residual attention U-Net), Model B (residual attention U-Net with Transformer bottleneck), and the Deep Transformer U-Net. Table I reports the best validation Dice and the corresponding test-set metrics after threshold tuning.

TABLE I
SEGMENTATION MODEL PERFORMANCE COMPARISON

Metric	Model A	Model B	Deep Trf. U-Net
Best Val Dice	0.878	0.878	0.868
Test Dice	0.896	0.897	0.889
Test IoU	0.812	0.813	0.800
Test Precision	0.887	0.887	0.871
Test Recall	0.905	0.907	0.908

All three models achieved test Dice scores close to 0.90 with IoU values around 0.80 and pixel accuracy above 0.98. Models A and B performed similarly, with Model B slightly ahead. The deeper Transformer variant underperformed despite greater depth and training time.

B. Effect of the Transformer Bottleneck

Comparing Model A and Model B isolates the impact of adding a Transformer bottleneck to the residual attention U-Net backbone. The hybrid model achieved a small improvement in test Dice and IoU and slightly smoother validation Dice curves, suggesting that limited global self-attention provides a modest benefit without changing overall precision and recall. As shown in Fig. 2, the validation Dice increases smoothly and plateaus around 0.88 without clear overfitting.

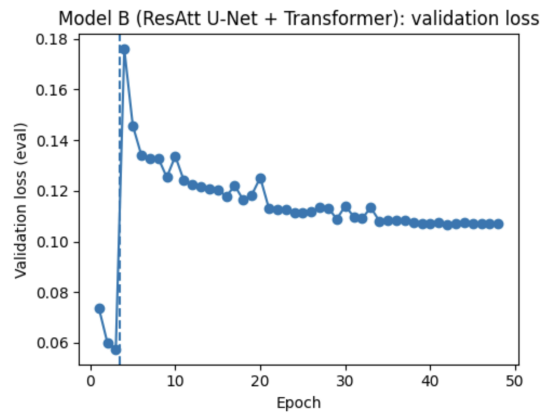

Fig. 1.

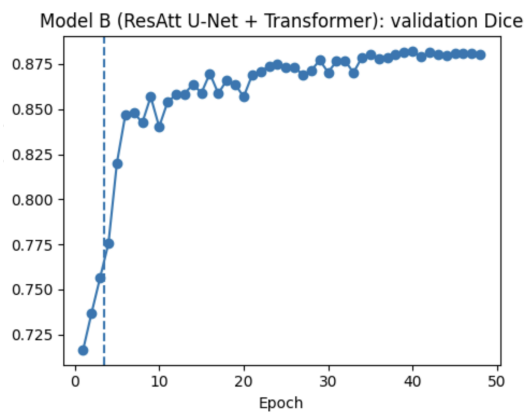

Fig. 2.

C. Impact of Depth and Edge-Enhanced Loss

The Deep Transformer U-Net increased both the depth of the encoder–decoder and the size of the Transformer bottleneck. Although it reached competitive validation Dice early in training, its best validation Dice (0.868) and test Dice (0.889) remained below those of the shallower hybrid model, indicating diminishing returns from further increasing capacity on this dataset.

IV. Discussion

In this study, the goal was to implement and evaluate the effectiveness of different U-net-based architectures for 2D single-cell segmentation on the provided baby hamster kidney cell dataset. From the data above, we can confirm the U-Net's strong performance in biomedical image segmentation, with all three variants achieving high accuracy metrics—Test Dice just below 0.90 and Test IoU values clustering around 0.81. The

two-stage training procedure was crucial in achieving these scores, clearing challenges like blurry edges and cell overlap.

Upon comparing the results of Model A (ResAtt U-Net) to Model B (ResAtt U-Net + Transformer), there is a notable improvement in Test Dice (A: 0.896 to B: 0.897) and IoU (A: 0.812 to B: 0.813). This suggests that Model B's incorporation of a lightweight Transformer bottleneck provides a slight benefit to the accuracy of the model. It is likely that this addition enhances the model's ability to capture global dependencies across the entire image, which is vital for accurately separating tightly clustered cells. This establishes the hybrid Model B architecture as an effective approach for this segmentation task.

Conversely, the Deep Transformer U-Net performed marginally worse (Test Dice: 0.889), possibly indicating that the model's depth and complexity did not translate to better performance on the given dataset. This is likely suggesting that the capacity of shallower models was already sufficient for the data's complexity, illustrating the concept of diminishing returns for architectural scaling.

The implications of Model B to perform consistently at near 90% accuracy opens the door to a powerful and verified tool to significantly accelerate the quantitative analysis of cell imagery in biological research. However, our study is subject to several limitations that need to be resolved before use in generalized clinical or research settings. First, the models were trained exclusively on a single dataset, which prevents generalizability to different cell types, tissues, or modalities without further validation. Second, the underperformance of the deepest model may suggest our training procedure, while robust for shallower models, may not have been fully optimized to the more complex Deep Transformer U-Net. As a result, future work should be done on cross-dataset validation on Model B to confirm its robustness, optimizing the Transformer component to improve efficiency, and finally, model deployment and optimization to enable its integration into real-time biological processing systems.

REFERENCES

[1] Litjens G, Kooi T, Bejnordi B E, et al. "A survey on deep learning in medical image analysis"[J]. Medical image analysis, 2017, 42: 60-88. https://arxiv.org/pdf/1702.05747.pdf

[2] Chartrand G, Cheng P M, Vorontsov E, et al. "Deep learning: a primer for radiologists"[J]. Radiographics, 2017, 37(7): 2113-2131. https://www.ncbi.nlm.nih.gov/pubmed/29131760

[3] Lundervold A S, Lundervold A. "An overview of deep learning in medical imaging focusing on MRI"[J]. Zeitschrift für Medizinische Physik, 2019, 29(2): 102-127 https://www.sciencedirect.com/science/article/pii/S0939388918301181

[4] Rajkomar A, Dean J, Kohane I. "Machine learning in medicine"[J]. New England Journal of Medicine, 2019, 380(14): 1347-1358. https://www.nejm.org/doi/pdf/10.1056/NEJMra1814259

[5] U-Net: Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. "U-net: Convolutional networks for biomedical image segmentation"[J]. International Conference on Medical image computing and computer-assisted intervention. Springer, Cham, 2015. https://link.springer.com/chapter/10.1007/978-3-319-24574-4_28

[6] LeCun Y, Bottou L, Bengio Y, et al. "Gradient-based learning applied to document recognition"[J]. Proceedings of the IEEE, 1998, 86(11): 2278-2324 https://ieeexplore.ieee.org/document/726791